

Math727 – Numerical Analysis II

Final Exam

1. Consider the two-point boundary problem

$$u'' = f(x, u, u'), \quad u(a) = \alpha, \quad u(b) = \beta$$

where $f : [a, b] \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is differentiable and satisfies that

$$0 < \frac{\partial f}{\partial u}(x, v_1, v_2) \leq K, \quad \frac{\partial f}{\partial u'}(x, v_1, v_2) \leq L, \quad (x, v_1, v_2) \in [a, b] \times \mathbb{R}^2,$$

for certain constants K, L . For $F(s) = u(b, s) - \beta$ that corresponds to the simple shooting method, show that for any $s \in \mathbb{R}$, it holds that

$$0 < k_1 \leq F'(s) \leq k_2$$

where

$$k_1 := \frac{1}{L}(1 - e^{-L(b-a)}), \quad k_2 := \frac{2e^{L\frac{b-a}{2}}}{C} \sinh\left(C\frac{b-a}{2}\right)$$

with $C = L\sqrt{1 + \frac{4K}{L^2}}$.

2. A square matrix $\mathbf{A} \in \mathbb{C}^{n \times n}$ is said to be *skew Hermitian* if $\mathbf{A}^H = -\mathbf{A}$. Prove that

- (a) The eigenvalues of a skew Hermitian are purely imaginary.
- (b) $\mathbf{I} - \mathbf{A}$ is non-singular.
- (c) $\mathbf{C} = (\mathbf{I} - \mathbf{A})^{-1}(\mathbf{I} + \mathbf{A})$ is unitary.

3. For the given matrices

$$\mathbf{A} = \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix} \in \mathbb{C}^{n \times n}$$

and $\mathbf{B} \in \mathbb{C}^{n \times n}$, let $\lambda_j(\theta)$, $j = 1, \dots, n$, denote the eigenvalues of the matrix $\mathbf{A} + \theta\mathbf{B}$ with $\theta \in \mathbb{C}$. Show that

- (a)

$$\max_{1 \leq j \leq n} |\lambda_j(\theta) - \lambda| \leq (\|\mathbf{B}\|_\infty + 1)|\theta|^{1/n}$$

for $|\theta| < 1$.

- (b) The estimate in (a) can not be improved in terms of the exponent $1/n$ of θ .

4. For any $\mathbf{x} \in \mathbb{R}^n$, define the function

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{b}^T \mathbf{x}$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a spd matrix and $\mathbf{b} \in \mathbb{R}^n$ is a vector. Show that for the *Steepest Descent* method applied to minimizing f the following inequality holds:

$$f(\mathbf{x}^{(k+1)}) \leq \left(\frac{\kappa_2(A) - 1}{\kappa_2(A) + 1} \right)^2 f(\mathbf{x}^{(k)})$$

5. To solve the following nonlinear system

$$\begin{cases} -\frac{1}{81} \cos x_1 + \frac{1}{9} x_2^2 + \frac{1}{3} \sin x_3 = x_1 \\ \frac{1}{3} \sin x_1 + \frac{1}{3} \cos x_3 = x_2 \\ -\frac{1}{9} \cos x_1 + \frac{1}{3} x_2 + \frac{1}{6} \sin x_3 = x_3 \end{cases}$$

use the fixed-point iteration $\mathbf{x}^{(k+1)} = \Phi(\mathbf{x}^{(k)})$ where $\Phi(\mathbf{x})$ is the left-hand side of the system.

- Analyze theoretically the convergence of the iterations to the fixed point $\mathbf{x}^* = (0, 1/3, 0)^T$.
 - Write a program to numerically compute an approximate solution with stopping criterion $\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|_\infty < 10^{-8}$ and the initial guess $\mathbf{x}^{(0)} = (1/2, 1/2, 1/2)$. Print out the results and draw a figure for the evolution of errors along iterations.
6. Given a nonsingular matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, its inverse $\mathbf{X} = \mathbf{A}^{-1}$ is obviously a solution of the nonlinear equation $\mathbf{A}\mathbf{X} - \mathbf{I} = \mathbf{0}$. Using Newton's method to solve it leads to the following iterations:

$$\mathbf{X}^{(k+1)} = \mathbf{X}^{(k)} + \mathbf{X}^{(k)}(\mathbf{I} - \mathbf{A}\mathbf{X}^{(k)}), \quad k = 0, 1, \dots$$

Show that for every initial matrix $\mathbf{X}^{(0)} \in \mathbb{R}^{n \times n}$ with $\|\mathbf{I} - \mathbf{A}\mathbf{X}^{(0)}\| \leq q < 1$ (for some vector-derived matrix norm $\|\cdot\|$), the sequence $\mathbf{X}^{(0)}, \mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \dots \subset \mathbb{R}^{n \times n}$ converges to the matrix $\mathbf{A}^{-1}$ with the inequality

$$\|\mathbf{X}^{(k)} - \mathbf{A}^{-1}\| \leq \frac{\|\mathbf{X}^{(0)}\|}{1 - q} \|\mathbf{I} - \mathbf{A}\mathbf{X}^{(k)}\| \leq \frac{\|\mathbf{X}^{(0)}\|}{1 - q} q^{(2^k)}$$

for $k = 0, 1, \dots$.